

WEEK-5

Task 1-Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>
#include <time.h>

int main() {
    int fd[2];
    pid_t pid;

    char buffer[100];
    char *message = "Data produced by parent process";

    struct timespec start, end;

    if (pipe(fd) == -1) {
        perror("pipe");
        return 1;
    }

    clock_gettime(CLOCK_MONOTONIC, &start);

    pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid > 0) {
        // Parent = Producer
        close(fd[0]);

        write(fd[1], message, strlen(message) + 1);

        close(fd[1]);
    }
}
```

```

wait(NULL);

clock_gettime(CLOCK_MONOTONIC, &end);

double time_taken =
    (end.tv_sec - start.tv_sec) * 1e6 +
    (end.tv_nsec - start.tv_nsec) / 1e3;

printf("Communication time: %.2f microseconds\n", time_taken);
}
else {
    // Child = Consumer
    close(fd[1]);

    read(fd[0], buffer, sizeof(buffer));

    printf("Child received: %s\n", buffer);

    close(fd[0]);
}

return 0;
}

```

```

ramya@LAPTOP-OPDAF309: ~$ mkdir prac5
ramya@LAPTOP-OPDAF309: ~$ cd prac5
ramya@LAPTOP-OPDAF309: ~/prac5$ nano task1.c
ramya@LAPTOP-OPDAF309: ~/prac5$ gcc task1.c -o task1
ramya@LAPTOP-OPDAF309: ~/prac5$ ./task1
Child received: Data produced by parent process
Communication time: 471.96 microseconds
ramya@LAPTOP-OPDAF309: ~/prac5$ |

```

Task 2-Develop a program that executes the equivalent of the shell command: `ls -l grep` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    int fd[2];
    pid_t p1, p2;

    if (pipe(fd) == -1) {
        perror("pipe");
        return 1;
    }

    // First child: executes ls -l
    p1 = fork();

    if (p1 < 0) {
        perror("fork");
        return 1;
    }

    if (p1 == 0) {
        close(fd[0]);

        dup2(fd[1], STDOUT_FILENO);
        close(fd[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("execlp ls");
        exit(1);
    }

    // Second child: executes grep txt
    p2 = fork();
```

```

    if (p2 < 0) {
        perror("fork");
        return 1;
    }

    if (p2 == 0) {
        close(fd[1]);

        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);

        execlp("grep", "grep", "txt", NULL);

        perror("execlp grep");
        exit(1);
    }

    // Parent closes both pipe ends
    close(fd[0]);
    close(fd[1]);

    waitpid(p1, NULL, 0);
    waitpid(p2, NULL, 0);

    return 0;
}

```

```

ramya@LAPTOP-OPDAF309:~/prac5$ ls -l | grep txt
ramya@LAPTOP-OPDAF309:~/prac5$ nano task2.c
ramya@LAPTOP-OPDAF309:~/prac5$ gcc task2.c -o task2
ramya@LAPTOP-OPDAF309:~/prac5$ ./task2
ramya@LAPTOP-OPDAF309:~/prac5$ touch test.txt
ramya@LAPTOP-OPDAF309:~/prac5$ ls -l
total 44
-rwxr-xr-x 1 ramya ramya 16392 Sep 16 06:50 task1
-rw-r--r-- 1 ramya ramya 1082 Sep 16 06:48 task1.c
-rwxr-xr-x 1 ramya ramya 16296 Sep 16 06:57 task2
-rw-r--r-- 1 ramya ramya 986 Sep 16 06:57 task2.c
-rw-r--r-- 1 ramya ramya 0 Sep 16 06:59 test.txt
ramya@LAPTOP-OPDAF309:~/prac5$ ./task2
-rw-r--r-- 1 ramya ramya 0 Sep 16 06:59 test.txt
ramya@LAPTOP-OPDAF309:~/prac5$ |

```